ITT06105 • DESKTOP PUBLISHING

IMAGE PROCESSING AND COMPRESSION

Lecture 10 | Ordinary Diploma in IT

Process → export → measure → inspect → choose

Academic year 2026/2027

Learning outcomes

After the lecture, you should be able to:

1	**Explain**	Describe image processing and its main steps.
2	**Distinguish**	Tell cropping, resizing and compression apart.
3	**Compare**	Explain lossy, lossless and common file formats.
4	**Calculate**	Find file-size percentage reduction and compression ratio.
5	**Recommend**	Select suitable images for print and digital use.

COURSE CONNECTION

Our publication needs images

Recall the “Stay Safe Online” hand-out created in previous lectures.

One picture, different destinations

An image may be printed in an A4 educational hand-out, shared through a phone or placed in a poster. The required dimensions, detail, transparency and file size will not always be identical. Image processing prepares the image for each job.

The right question is not “Which export is smallest?” but “Which export meets the intended use?”

KEY CONCEPT

What is image processing?

A foundational term for the final topic.

Definition

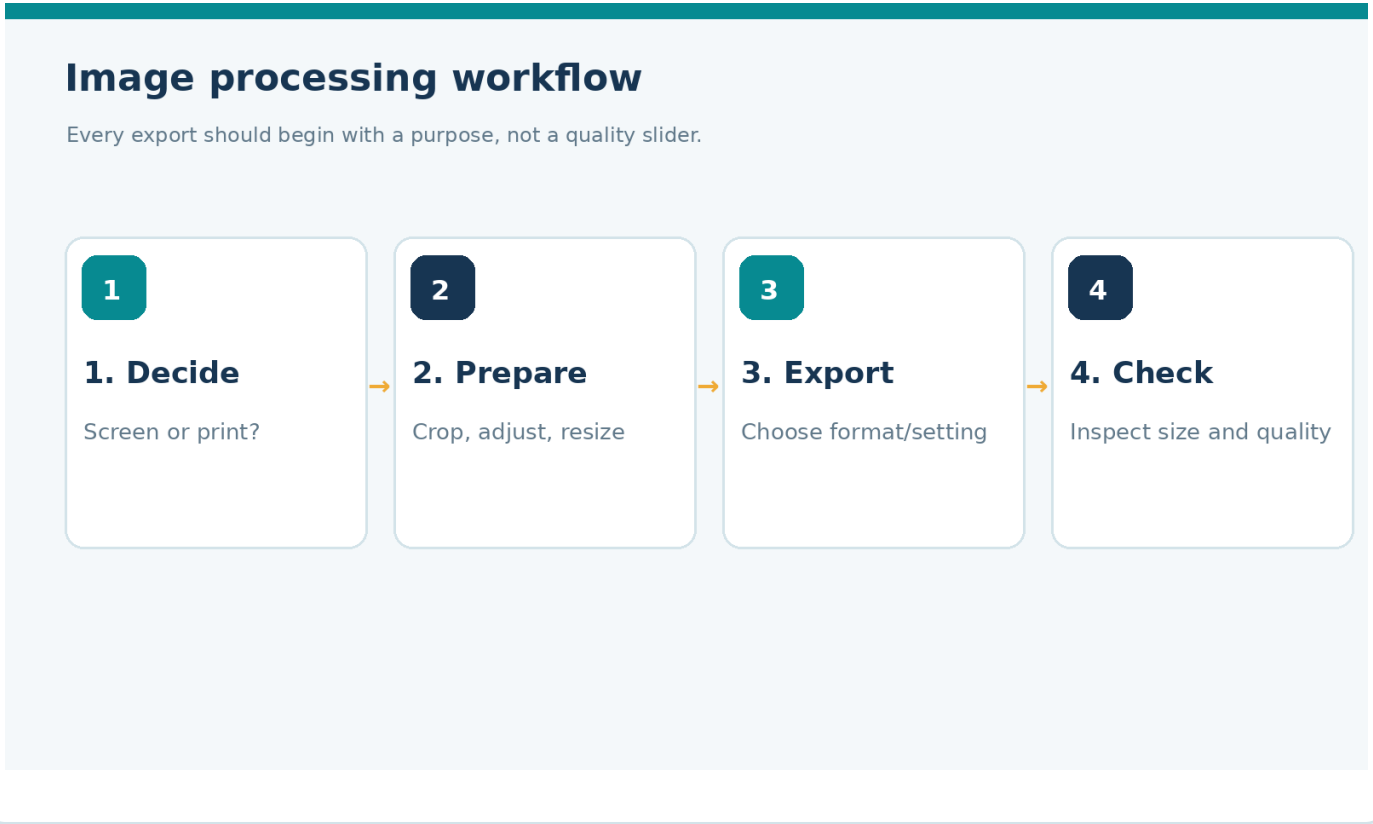
Image processing is the use of operations to modify, improve, analyse or prepare a digital image for a particular purpose. In desktop publishing, common operations include cropping, resizing, correcting brightness, adjusting contrast and preparing export files.

For example, an image may be cropped to remove distractions before placement in the hand-out.

ILLUSTRATION

A simple image-processing workflow

Four stages help avoid repeated trial and error.



Start with a purpose

Decide where the image will be used. Make necessary corrections; export a suitable format and setting; then inspect the result at the intended size. If it fails, adjust the source or export rather than accepting a poor output.

EXPLAIN

Three common processing operations

These actions are related, but not identical.

Cropping

Cropping removes unwanted outer parts of a picture. The remaining image may have a different shape or composition. A crop cannot restore details already outside the selected area.

Resizing and correction

Resizing changes pixel dimensions, such as 1,200 × 800 to 600 × 400. Correction changes appearance, for example exposure, contrast or colour. Both may be necessary before exporting.

ILLUSTRATION

Resizing versus compression

File size can change for different reasons.

Resizing is not the same as compression

Changing dimensions removes pixels; changing format or quality changes how data are stored.

Original

1,200 × 800 px



Resized

600 × 400 px



Same pixels, new encoding

1,200 × 800 px



Distinguish the changes

Resizing reduces or increases the pixel grid. Compression changes how image information is encoded. Two exports can have exactly the same pixel dimensions but different file sizes. A fair comparison needs both dimensions and file sizes recorded.

KEY CONCEPT

Pixels and image dimensions

Raster images store colour at positions in a pixel grid.

Pixel dimensions

A raster picture measuring 1,200 × 800 contains 1,200 pixels horizontally and 800 vertically. Those numbers describe pixel count, not the physical width in centimetres or the size of the stored file in KiB.

If we double its printed width without adding detail, its effective PPI becomes lower.

KEY CONCEPT

Cropping as a design decision

A crop should strengthen the message without hiding important information.

What the Crop Tool changes

Cropping removes areas outside the chosen region. It can improve emphasis and make a photograph fit a text-and-image layout. However, removing part of a person, object or caption may change the meaning. Always check that essential content remains visible.

Example: crop a classroom picture to show the learner and computer more clearly.

Basic image preparation

Use a beginner-friendly workflow in Photoshop or GIMP.

1	**Open safely**	Work from a copy, keeping the original untouched.
2	**Crop**	Choose a suitable aspect ratio and keep meaningful content.
3	**Resize if needed**	Set sensible pixel dimensions without distorting proportions.
4	**Adjust lightly**	Improve contrast or brightness only when the result is better.
5	**Export and review**	Save the source, then inspect the exported image.

Choose the correct operation

Consider what the student wants to change.

YOUR TURN

A photo is 1,200 × 800 pixels. Its subject is too small, and the image is too large to upload. Which two separate operations might help?

Explain the goal of cropping and the goal of export compression separately.

KEY CONCEPT

What does image compression do?

Compression is about storing and distributing image information.

Definition

Image compression reduces the amount of digital storage required to represent an image. A compressed image may upload faster or require less space. Compression can be lossless or lossy, and the visible result depends on the format, settings and image content.

Compression should not be confused with simply changing image dimensions.

ILLUSTRATION

Two main kinds of compression

Know the difference before choosing a file format.

Lossless versus lossy

They reduce storage in different ways.

LOSSLESS

Keeps image information so the file can be reconstructed without lost data.

PNG often suits icons, screenshots and graphics with transparent areas.

LOSSY

Discards some information, often achieving smaller files.

JPEG and lossy WebP are common for photographs and web delivery.

Two strategies

Lossless methods preserve all image information needed for exact reconstruction of the encoded image. Lossy methods discard selected information to produce potentially smaller outputs, but may harm fine details.

KEY CONCEPT

Lossless compression

Useful when faithful detail or crisp graphic edges matter.

Basic idea

A lossless format can recover the stored pixels without data being removed by the compression process. PNG uses lossless compression and is commonly chosen for screenshots, graphic elements and images that require transparency.

Lossless does not mean every PNG is small; complicated images can still have large files.

KEY CONCEPT

Lossy compression

Smaller files can involve a visible quality trade-off.

Basic idea

Lossy compression discards some image information. JPEG is a common lossy format. When compression is stronger, the file may become smaller, but smooth areas, fine lines and text may show blocks, halos or loss of detail.

Lossy image quality should be judged by looking at the actual export at a suitable viewing size.

ILLUSTRATION

Compare image details

The practical uses four exports with equal pixel dimensions.

Compare the same area

Look for blocked detail, jagged edges and softness at equal viewing size.



Original PNG

Reference export



JPEG Q90

Same 1200×800 source dimensions



JPEG Q55

Same 1200×800 source dimensions



JPEG Q25

Same 1200×800 source dimensions

Inspect fairly

Each version is based on the same 1,200 × 800 image. Compare the same region at the same displayed scale; then view the file as it would appear in the final publication. Compression differences may be clearer in some regions than others.

EXPLAIN

What does JPEG quality mean?

A useful setting, but not a universal numerical measurement.

Higher setting

A high-quality JPEG setting usually retains more visual detail but tends to produce a larger file. A photographic announcement may benefit from this when fine edges or colour transitions matter.

Lower setting

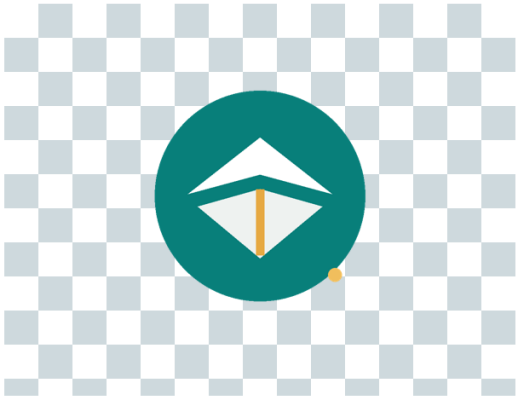
A lower setting often gives a smaller file, but may create visible artefacts. Settings such as Q90 and Q55 are encoder controls, not a universal claim that the resulting image retains exactly 90% or 55% of its information.

Transparency needs attention

The background can be part of a graphic requirement.

Transparency changes the decision

A transparent icon needs an output format that preserves an alpha channel.



Keep transparency

PNG preserves transparent regions. A JPEG export does not; it usually fills them with a background colour.

PNG and SVG can help

A transparent badge can sit over different page colours. PNG preserves raster transparency, while SVG can keep a vector icon scalable. Ordinary JPEG does not preserve a transparency channel, making it a poor choice for an icon that needs a clear background.

ILLUSTRATION

Common image formats

A quick comparison of basic publishing choices.

Which format should I choose?

Make the choice from the content and the delivery requirements.

JPEG

Photographs

Usually lossy; no transparency

PNG

Text/graphics

Lossless; supports alpha

WebP

Web images

Lossy or lossless options

TIFF

Professional workflows

Large files; print interchange

Match format to content

JPEG is often used for photographs. PNG is useful for crisp graphics or transparency. WebP supports efficient web exports when compatible with the delivery platform. TIFF may appear in specialist print workflows; follow the print provider’s requirements.

EXPLAIN

JPEG and PNG in practice

Both may be used in the same publication.

JPEG — photographs

Usually effective for natural photographic images with many colours and gradients. It does not support alpha transparency, and repeated lossy re-saving should be avoided when an editable source is available.

PNG — graphic elements

Often useful for flat-colour graphics, screenshots, diagrams and transparent icons. It stores pixels without lossy compression, though photographic PNG files may be much larger than suitable JPEG exports.

WebP, TIFF and source files

Not every file is intended for the same workflow.

WebP and TIFF

WebP can provide efficient delivery for websites and supported applications; it offers lossy and lossless modes. TIFF is commonly encountered in some high-quality image and print exchanges and may generate larger files.

Keep editable source files

A Photoshop PSD is a working project with layers, while an exported JPEG or PNG is intended for delivery. Preserve the editable version so later changes do not require repeatedly degrading an exported image.

Choose a format

A good answer explains why the format fits the purpose.

YOUR TURN

Would you normally use a transparent PNG or a JPEG to place a lightbulb icon over a coloured hand-out background? Explain your choice.

You may also mention an SVG if an editable vector icon is supported.

KEY CONCEPT

How do we measure file size?

Two units you may encounter on a computer.

Bytes and kibibytes

A byte is a unit of stored digital information. One kibibyte (KiB) is 1,024 bytes. File size and image dimensions are different measurements; two 1,200 × 800 images may have very different file sizes because of encoding and content.

The practical uses KiB to describe the files measured from its supplied image set.

KEY FORMULA

Percentage file-size reduction

A simple calculation describes how much smaller an export becomes.

Formula

Percentage reduction = ((Original size − New size) ÷ Original size) × 100. Use the same size unit on both sides and make sure the new size is smaller. A reduction of 75% means that one-quarter of the original file size remains.

Example: original 800 KiB, new 200 KiB → (800 − 200) ÷ 800 × 100 = 75%.

WORKED EXAMPLE

Worked example: 800 KiB to 200 KiB

Write the units and show your arithmetic.

Calculation

Original file O = 800 KiB; exported file N = 200 KiB. Difference = 600 KiB. Percentage reduction = (600 ÷ 800) × 100 = 75%. The export is 25% of the original size.

A smaller file does not prove that the visual result remains acceptable.

ILLUSTRATION

Real measurements from our files

The class can verify these sizes with a file manager.

Example measurements

These values are measured from the supplied classroom files.

FILE	PIXELS	SIZE	REDUCTION
Original	1200×800	712.8 KiB	0.0%
JPEG_Q90	1200×800	159.4 KiB	77.6%
JPEG_Q55	1200×800	47.7 KiB	93.3%
JPEG_Q25	1200×800	32.0 KiB	95.5%
WebP_Q70	1200×800	20.6 KiB	97.1%

Interpret the table

The original PNG is approximately 712.8 KiB. JPEG Q55 is approximately 47.7 KiB, a reduction of about 93.3%. These measured files all have 1,200 × 800 pixel dimensions. Another encoder may produce different values.

WORKED EXAMPLE

Worked example: PNG versus JPEG

Use measured bytes or rounded KiB consistently.

Approximate reduction

Original PNG = 712.8 KiB. JPEG Q55 = 47.7 KiB. Reduction = (712.8 − 47.7) ÷ 712.8 × 100 ≈ 93.3%. The JPEG file occupies roughly 6.7% of the PNG size.

Check readability before recommending the JPEG simply because it is smaller.

KEY FORMULA

What is a compression ratio?

Another way to express the size change.

Original size : New size

An original-to-new compression ratio describes how many units of old storage correspond to one unit of new storage. For 800 KiB reduced to 200 KiB, the ratio is 800:200, or 4:1.

A 4:1 ratio is not the same number as the 75% reduction, though they describe the same size change.

Calculate with a colleague

Try the calculation before checking the solution.

YOUR TURN

A file of 600 KiB becomes 150 KiB. What are the percentage reduction and original-to-new compression ratio?

Expected reasoning: calculate the difference first, then divide by the original.

KEY CONCEPT

Make fair comparisons

Do not change too many variables at once.

Control pixel dimensions

To compare export settings, begin with one original image and keep pixel dimensions constant. Record each format, quality setting and resulting file size. A 600 × 400 resized image should be analysed separately from the 1,200 × 800 exports because pixel count has also changed.

The smallest file in our practice set is not necessarily the preferred print image.

ILLUSTRATION

What should be checked visually?

View exports at equal size and at their real intended display size.

A simple image-quality checklist

Inspect at both equal displayed size and intended use size.

Text

Are letters readable?

Edges

Are there jagged or blocky outlines?

Colour

Are colours acceptable for the job?

Transparency

Is any needed clear background retained?

A practical inspection

Look at text, edges, colour, photographic detail and transparency where needed. Strong compression artefacts can make a graphic hard to read, even if the file transfers quickly. Notice whether the output still communicates its intended message.

KEY CONCEPT

What are compression artefacts?

Sometimes lost detail appears as visible defects.

Recognising common problems

At strong lossy settings, an image may show square blocks, fuzzy edges, ringing around high-contrast lines or blurred small lettering. Artefacts are more noticeable in some types of content than others. Compare both the detailed view and normal viewing size.

A clean high-resolution source will not be restored by increasing JPEG export quality after it has already been damaged.

COURSE CONNECTION

Image quality for printing

Recall effective PPI from Lecture 9.

Effective PPI

Image pixel density at the final physical print size can affect clarity. Effective PPI = (pixel width × 25.4) ÷ printed width in millimetres. This is not the same as printer DPI. The required image quality depends on the print context and provider.

An image that looks fine on a phone may not be detailed enough when printed much larger.

WORKED EXAMPLE

Worked print example

Link image dimensions to the size on the page.

1,200 pixels across 150 mm

Effective PPI = 1,200 × 25.4 ÷ 150 = 203.2 PPI. This tells us the pixel density at the intended printed width. It is one factor in print suitability, not a universal guarantee of acceptable output.

Ask the printer for any required image-resolution specification.

ILLUSTRATION

Choosing images for three uses

A suitable format depends on the job.

The best file is fit for purpose

An efficient image that is unreadable or lacks transparency is not a successful export.

Phone announcement

Consider JPEG or WebP

Readability and reasonable file size

Printable hand-out

Keep sufficient pixels

Check effective PPI at final placement

Transparent learning icon

Choose PNG or SVG

Transparent background and sharp edges

Three different decisions

A mobile announcement may favour a compact JPEG or supported WebP. A printed hand-out needs suitable image detail at final output size. A transparent icon can be kept as SVG or exported to a PNG that preserves transparent areas.

Our practical exercise

Use the provided images and measurement sheet.

1	**Open and compare**	Review original PNG, JPEG and WebP exports.
2	**Record measurements**	Note dimensions, KiB values and settings.
3	**Calculate**	Find a percentage reduction and compression ratio.
4	**Inspect quality**	Look for artefacts at equal viewing sizes.
5	**Recommend**	Select suitable images for phone, print and icon use.

What students should submit

A short record of evidence is enough for this beginner exercise.

Minimum student output

Submit the completed practical worksheet, identified file versions and a brief explanation of the trade-off between quality and file size. If software access is limited, use the prepared exports and measurements to complete the analysis.

When you can access Photoshop or GIMP, repeat the export process using your own saved settings.

KEY TAKEAWAY

Key lesson: small is not always better

An export is successful when it serves the intended audience.

Summary

Image processing includes preparing, adjusting and exporting images. Resizing changes pixels; compression changes stored information. Lossless and lossy methods have different trade-offs. Record file sizes, dimensions and visual quality before choosing an output.

Keep editable sources, label your exports and choose file formats for their actual purpose.

THE COMPLETE DTP WORKFLOW

Plan → create → arrange → print → optimise

Choose an image format by considering image quality, file size and the intended use.